

# Foundations of Image captioning

[Image captioning]

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## 1.0 Content Block

Chatbots Another area where NLG has been widely applied is automated dialogue systems, frequently in the form of chatbots. A chatbot or chatterbot is a software application used to conduct an on-line chat conversation via text or text-to-speech, in lieu of providing direct contact with a live human agent. While natural language processing (NLP) techniques are applied in deciphering human input, NLG informs the output part of the chatbot algorithms in facilitating real-time dialogues. Early chatbot systems, including Cleverbot created by Rollo Carpenter in 1988 and published in 1997, reply to questions by identifying how a human has responded to the same question in a conversation database using information retrieval (IR) techniques. Modern chatbot systems predominantly rely on machine learning (ML) models, such as sequence-to-sequence learning and reinforcement learning to generate natural language output. Hybrid models have also been explored. For example, the Alibaba shopping assistant first uses an IR approach to retrieve the best candidates from the knowledge base, then uses the ML-driven seq2seq model re-rank the candidate responses and generate the answer.

## 2.0 Content Block

Pollen counts are expected to remain high at level 6 over most of Scotland, and even level 7 in the south east. The only relief is in the Northern Isles and far northeast of mainland Scotland with medium levels of pollen count.

## 3.0 Content Block

Natural language generation (NLG) is a software process that produces natural language output. A widely cited survey of NLG methods describes NLG as "the subfield of artificial intelligence and computational linguistics that is concerned with the construction of computer systems that can produce understandable texts in English or other human languages from some underlying non-linguistic representation of information". While it is widely agreed that the output of any NLG process is text, there is some disagreement about whether the inputs of an NLG system need to be non-linguistic. Common applications of NLG methods include the production of various reports, for example weather and patient reports; image captions; and chatbots like ChatGPT. Automated NLG can be compared to the process humans use when they turn ideas into writing or speech. Psycholinguists prefer the term language production for this process, which can also be described in mathematical terms, or modeled in a computer for psychological research. NLG systems can also be compared to translators of artificial computer languages, such as decompilers or transpilers, which also produce human-readable code generated from an intermediate representation. Human languages tend to be considerably more complex and allow for much more ambiguity and variety of expression than programming languages, which makes NLG more challenging. NLG may be viewed as complementary to natural language understanding (NLU): whereas in natural-language understanding, the system needs to disambiguate the input sentence to produce the machine representation language, in NLG the system needs to make decisions about how to put a representation into words. The practical considerations in building NLU vs. NLG systems are not symmetrical. NLU needs to deal with ambiguous or erroneous user input, whereas the ideas the system wants to express through NLG are generally known precisely. NLG needs to choose a specific, self-consistent textual representation from many potential representations, whereas NLU generally tries to produce a single, normalized representation of the idea expressed. NLG has existed since ELIZA was developed in the mid 1960s, but the methods were first used commercially in the 1990s. NLG techniques range from simple template-based systems like a mail merge that generates form letters, to systems that

have a complex understanding of human grammar. NLG can also be accomplished by training a statistical model using machine learning, typically on a large corpus of human-written texts.

#### 4.0 Content Block

**Stages** The process to generate text can be as simple as keeping a list of canned text that is copied and pasted, possibly linked with some glue text. The results may be satisfactory in simple domains such as horoscope machines or generators of personalized business letters. However, a sophisticated NLG system needs to include stages of planning and merging of information to enable the generation of text that looks natural and does not become repetitive. The typical stages of natural-language generation, as proposed by Dale and Reiter, are:

#### 5.0 Content Block

**Evaluation** As in other scientific fields, NLG researchers need to test how well their systems, modules, and algorithms work. This is called evaluation. There are three basic techniques for evaluating NLG systems:

#### 6.0 Content Block

**Creative writing and computational humor** Creative language generation by NLG has been hypothesized since the field's origins. A recent pioneer in the area is Phillip Parker, who has developed an arsenal of algorithms capable of automatically generating textbooks, crossword puzzles, poems and books on topics ranging from bookbinding to cataracts. The advent of large pretrained transformer-based language models such as GPT-3 has also enabled breakthroughs, with such models demonstrating recognizable ability for creating-writing tasks. A related area of NLG application is computational humor production. JAPE (Joke Analysis and Production Engine) is one of the earliest large, automated humor production systems that uses a hand-coded template-based approach to create punning riddles for children. HAHAAcronym creates humorous reinterpretations of any given acronym, as well as proposing new fitting acronyms given some keywords. Despite progresses, many challenges remain in producing automated creative and humorous content that rival human output. In an experiment for generating satirical headlines, outputs of their best BERT-based model were perceived as funny 9.4% of the time (while real headlines from The Onion were 38.4%) and a GPT-2 model fine-tuned on satirical headlines achieved 6.9%. It has been pointed out that two main issues with humor-generation systems are the lack of annotated data sets and the lack of formal evaluation methods, which could be applicable to other creative content generation. Some have argued relative to other applications, there has been a lack of attention to creative aspects of language production within NLG. NLG researchers stand to benefit from insights into what constitutes creative language production, as well as structural features of narrative that have the potential to improve NLG output even in data-to-text systems.

#### 7.0 Content Block

**Image captioning** Over the past few years, there has been an increased interest in automatically generating captions for images, as part of a broader endeavor to investigate the interface between vision and language. A case of data-to-text generation, the algorithm of image captioning (or automatic image description) involves taking an image, analyzing its visual content, and generating a textual description (typically a sentence) that verbalizes the most prominent aspects of the image. An image captioning system involves two sub-tasks. In Image Analysis, features and attributes of an image are detected and labelled, before mapping these outputs to linguistic structures. Recent research utilizes deep learning approaches through features from a pre-trained convolutional neural network such as AlexNet, VGG or Caffe, where caption generators use an activation layer from the pre-trained network as their input features. Text Generation, the second task, is performed using a wide range of techniques. For example, in the Midge system, input images are represented as triples consisting of object/stuff detections, action/pose detections and spatial relations. These are subsequently mapped to <noun, verb, preposition> triples and realized using a tree substitution grammar. A common method in image captioning is to use a

vision model (such as a ResNet) to encode an image into a vector, then use a language model (such as an RNN) to decode the vector into a caption. Despite advancements, challenges and opportunities remain in image capturing research. Notwithstanding the recent introduction of Flickr30K, MS COCO and other large datasets have enabled the training of more complex models such as neural networks, it has been argued that research in image captioning could benefit from larger and diversified datasets. Designing automatic measures that can mimic human judgments in evaluating the suitability of image descriptions is another need in the area. Other open challenges include visual question-answering (VQA), as well as the construction and evaluation multilingual repositories for image description.